

Predicting Academic Performance Using Machine Learning and Explainable AI Techniques

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Abstract. Academic performance is influenced by various educational, behavioral, and psychological factors. This project investigates the use of machine learning techniques to predict academic performance using three independent datasets related to student behavior, gaming habits, and mental health conditions. A complete data science pipeline was developed, including data cleaning, feature engineering, exploratory data analysis, model training, evaluation, and explainability analysis. Six regression algorithms were implemented and compared: Linear Regression, Decision Tree, Random Forest, Support Vector Regression (SVR), K-Nearest Neighbors (KNN), and Gradient Boosting. Hyperparameter optimization was performed using GridSearchCV with 5-Fold Cross Validation, while model performance was evaluated using RMSE, MAE, R^2 , and Adjusted R^2 metrics. The results showed that ensemble-based models achieved the highest predictive performance. Gradient Boosting produced the best results for the Gaming and Student datasets, while Random Forest achieved the best relative performance among the evaluated models on the Mental Health dataset, although the overall predictive performance remained limited. The Student Dataset achieved the highest predictive performance with an R^2 score of 0.9521.

Keywords: Machine Learning, Academic Performance Prediction, Educational Data Mining, Explainable AI, SHAP, Feature Engineering.

1 Introduction

Academic success is one of the most important indicators of student development and future opportunities. Understanding the factors that influence academic performance can help educators, institutions, and policymakers develop strategies to improve educational outcomes.

Student performance is affected by various behavioral, social, and psychological factors. Variables such as study habits, gaming activities, sleep duration, attendance, and mental health conditions may all contribute to academic achievement. With the increasing availability of educational data, machine learning techniques have become valuable tools for identifying patterns and predicting student outcomes.

Recent advances in Educational Data Mining (EDM) and Learning Analytics have enabled researchers to analyze large volumes of student-related data and identify factors influencing academic achievement. Machine learning algorithms are particularly useful because they can capture complex relationships among academic, behavioral, and psychological variables that may not be easily detected through traditional statistical methods.

Many previous studies focus on a single aspect of student behavior. In contrast, this project evaluates academic performance using three different datasets representing educational, gaming, and mental health factors. Furthermore, the proposed framework integrates data preprocessing, feature engineering, model optimization, explainability analysis, and automated testing into a single reproducible machine learning pipeline.

The main objective of this study is to develop and evaluate machine learning models capable of predicting academic performance using different student-related datasets. In addition, the study aims to improve model interpretability through explainable artificial intelligence techniques. The specific objectives are listed below.

- To preprocess and prepare multiple educational datasets.
- To engineer meaningful features from raw variables.
- To perform exploratory data analysis and identify important patterns.
- To compare multiple machine learning regression algorithms.
- To evaluate model performance using standard regression metrics.
- To explain model predictions using Feature Importance, Permutation Importance, and SHAP analysis.

2 Datasets

Three datasets were used in this study, each targeting a different aspect of student life and academic performance.

2.1 Student Dataset

The Student Dataset contains information related to academic habits, attendance, study behavior, and demographic characteristics of students. The target variable is Final CGPA. The dataset includes demographic characteristics, attendance records, study habits, previous academic performance, sleep patterns, and social activity indicators. These variables provide a comprehensive representation of student learning behavior and academic engagement.

2.2 Gaming Dataset

The Gaming Dataset contains information regarding gaming behavior, study time, sleep patterns, and academic grades. The target variable is Grades. The dataset captures both academic and gaming-related behaviors, allowing the investigation of how gaming habits, screen exposure, and lifestyle factors may influence student performance.

2.3 Mental Health Dataset

The Mental Health Dataset contains information related to depression, anxiety, panic attacks, age, academic pressure, and academic performance. The target variable is Final CGPA. The dataset provides information about psychological well-being and educational background, enabling the analysis of potential relationships between mental health conditions and academic outcomes. Because the original Mental Health Dataset stored academic performance as CGPA intervals rather than exact numerical values, the midpoint of each interval was used to generate a continuous Final CGPA variable suitable for regression analysis.

2.4 Dataset Summary

Table 1 summarizes the characteristics of the datasets used in this study. The datasets were selected to represent different dimensions of student behavior, including academic habits, gaming activities, and psychological well-being.

The Student Dataset focuses on academic engagement and lifestyle variables. The Gaming Dataset investigates the relationship between gaming behavior and academic outcomes. The Mental Health Dataset examines psychological factors that may influence academic performance.

Using multiple datasets allows a broader evaluation of machine learning models and helps identify which types of variables provide stronger predictive signals for academic achievement.

Table 1. Dataset Summary

Dataset	Records	Features	Target Variable
Student Dataset	5000	17	Final CGPA
Gaming Dataset	8000	22	Grades
Mental Health Dataset	100	60	Final CGPA

3 Methodology

[Figure 1. Proposed Machine Learning Framework]

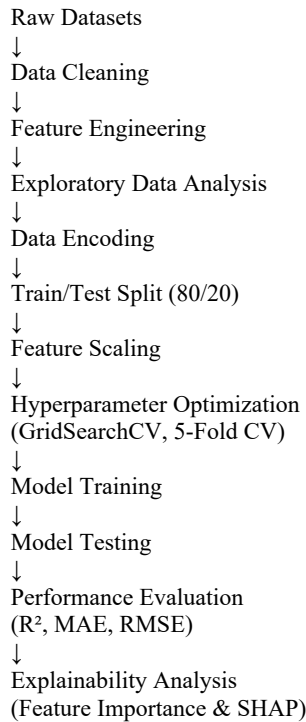


Figure 1 illustrates the overall machine learning workflow used in this study. The process begins with data cleaning and feature engineering, followed by exploratory data analysis and preprocessing. After splitting the data into training and testing sets, machine learning models are optimized using GridSearchCV with 5-Fold Cross Validation. The final models are evaluated using regression metrics and interpreted using explainable artificial intelligence techniques, including Feature Importance and SHAP analysis.

3.1 Data Cleaning

The raw datasets were cleaned before analysis. Missing values, inconsistent records, and formatting issues were identified and corrected. Categorical variables were standardized and prepared for machine learning processing. The cleaned datasets were stored in the processed data directory for further analysis. During data cleaning, rows containing missing values were removed from all datasets. Column names were standardized to maintain consistency across datasets, and several variables were renamed to create a unified schema. Categorical variables such as gender, marital status, depression, anxiety, panic attack, and treatment status were converted into numerical representations suitable for machine learning algorithms.

For the Mental Health Dataset, CGPA ranges were transformed into numerical CGPA values by calculating the midpoint of each interval. Year of study information was extracted and converted

into numerical values. Additional preprocessing steps included one-hot encoding of gender categories and the creation of standardized student identifiers. These transformations ensured compatibility across datasets and improved the quality of the machine learning pipeline.

3.2 Feature Engineering

Feature engineering was performed to create more meaningful variables from the original datasets. The purpose was to improve predictive performance and generate higher-level behavioral indicators. Examples of engineered features include

Gaming Intensity,

Academic Pressure Index,

Sleep-Gaming Balance,

Total Screen Time,

Total Mental Risk,

Study Efficiency,

Attendance Impact. These engineered variables were designed to capture interactions between existing features and provide a better representation of student behavior. In addition, they improve model interpretability by helping explain how different behavioral and academic factors contribute to prediction outcomes.

The engineered features were calculated using Eqs. (1)–(7).

$$\text{Gaming Intensity} = \text{Gaming Hours} / 24 \quad (1)$$

Eq. (1) defines the Gaming Intensity feature by measuring the proportion of daily time spent on gaming.

$$\text{Academic Pressure Index} = \text{Study Hours} / (\text{Gaming Hours} + 1) \quad (2)$$

Eq. (2) defines the Academic Pressure Index by representing the balance between study effort and gaming behavior. The addition of 1 prevents division by zero.

$$\text{Sleep-Gaming Balance} = \text{Sleep Hours} / (\text{Gaming Hours} + 1) \quad (3)$$

Eq. (3) defines the Sleep-Gaming Balance feature by capturing the relationship between sleep duration and gaming activity.

$$\text{Total Screen Time} = \text{Social Hours} + \text{Gaming Hours} \quad (4)$$

Eq. (4) defines Total Screen Time by combining social media usage and gaming hours to estimate overall digital exposure.

$$\text{Total Mental Risk} = \text{Depression} + \text{Anxiety} + \text{Panic Attack} \quad (5)$$

Eq. (5) defines Total Mental Risk by aggregating mental-health-related indicators into a single risk score.

$$\text{Study Efficiency} = \text{Study Hours} / ((\text{Social Hours} / 7) + 1) \quad (6)$$

Eq. (6) defines Study Efficiency by measuring effective study time relative to social media usage.

$$\text{Attendance Impact} = (\text{Attendance} \times \text{Previous CGPA}) / 100 \quad (7)$$

Eq. (7) defines Attendance Impact by combining attendance rate and previous academic performance.

The engineered features were created to improve predictive performance and enhance model interpretability. Rather than relying solely on raw variables, these features capture interactions between behavioral, academic, and psychological factors. For example, Academic Pressure Index reflects the balance between study effort and gaming activity, while Attendance Impact combines attendance behavior with prior academic achievement. Such transformations provide a more meaningful representation of student behavior and contribute to improved model performance.

3.3 Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed to better understand the datasets and identify relationships between variables. The following visualizations were generated: histograms, correlation heatmaps, boxplots, scatterplots, pairplots, and correlation ranking analysis. The analysis helped identify feature distributions, outliers, and potential relationships with academic performance indicators. Figures 2 to 5 present the main exploratory results for the Student dataset.

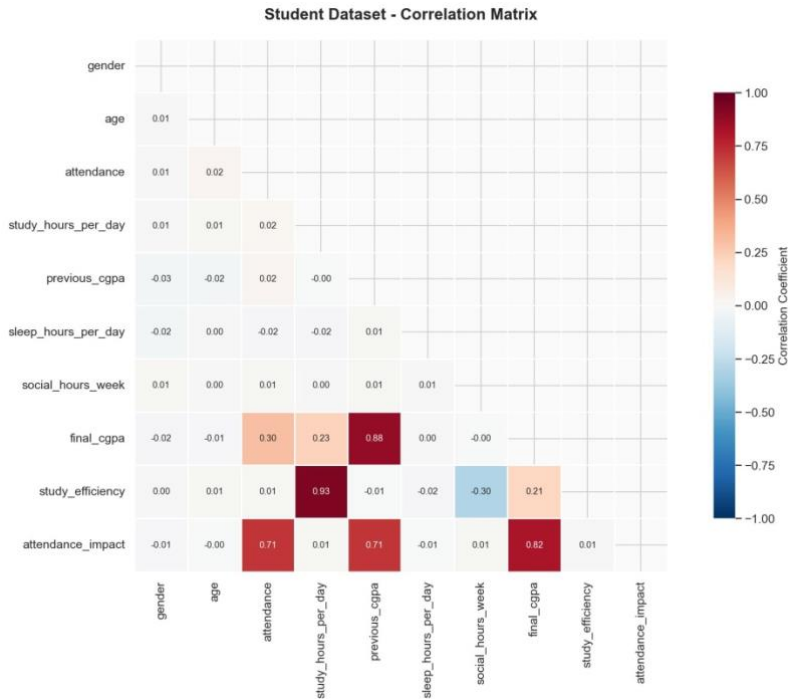


Fig. 2. Student Dataset Correlation Matrix.

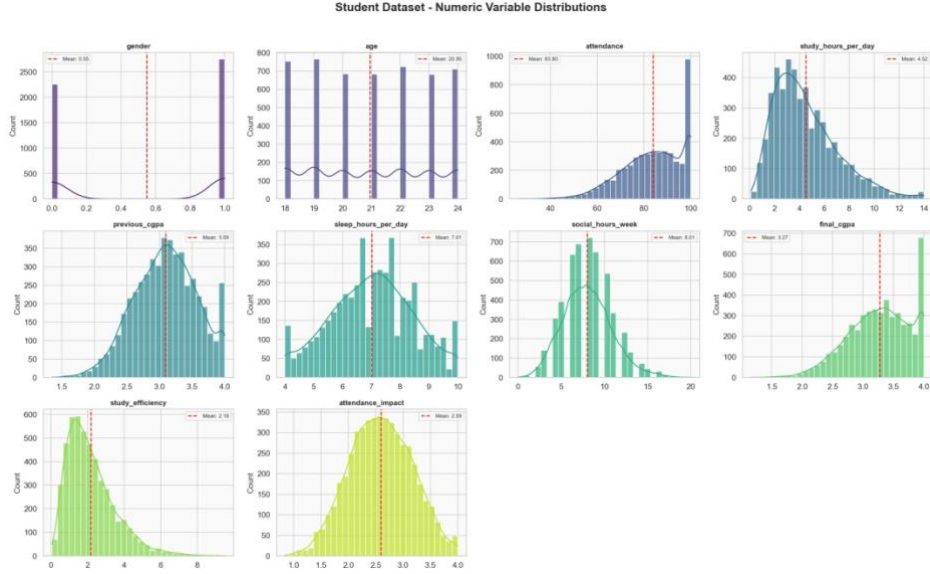


Fig. 3. Student Dataset Numeric Variable Distributions.

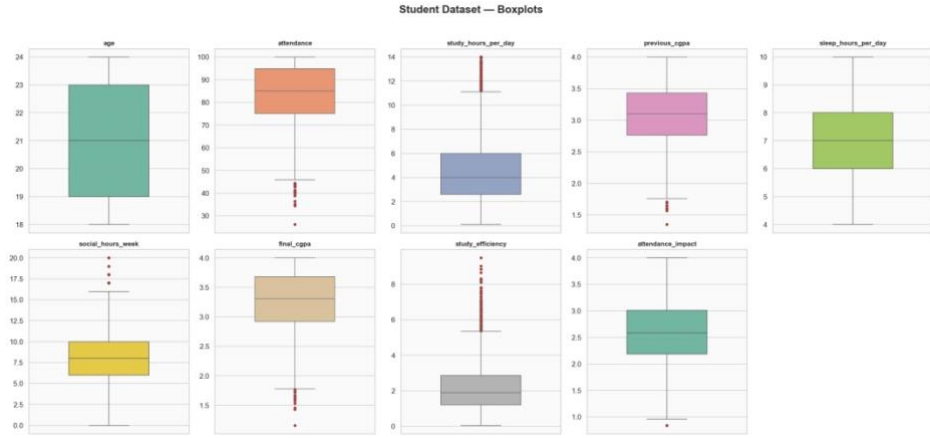


Fig. 4. Student Dataset Boxplots.

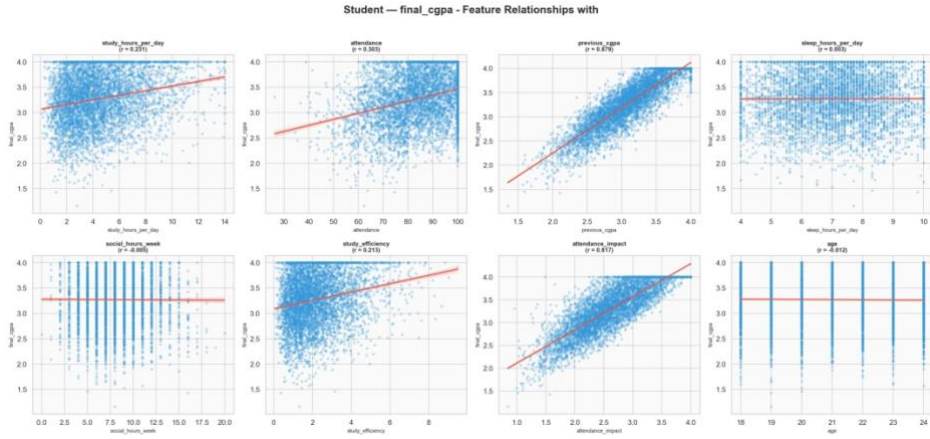


Fig. 5. Relationship Between Academic Features and Final CGPA.

The exploratory analysis revealed several meaningful patterns within the Student Dataset. Previous CGPA, attendance, and study efficiency exhibited positive relationships with academic performance indicators. Distribution analyses showed that most variables followed reasonable distributions, while boxplots identified several outliers that were retained because they represent realistic student behavior. The observed relationships suggest that academic engagement variables are important predictors of student success.

3.4 Machine Learning Pipeline

A complete machine learning pipeline was developed using Scikit-Learn. The pipeline included Train-Test Split, StandardScaler, Cross Validation, GridSearchCV Hyperparameter Optimization, and Model Evaluation. To prevent data leakage, StandardScaler was integrated directly into Scikit-Learn pipelines. This ensured that scaling operations were performed only on training data during cross-validation. Hyperparameter optimization was conducted using GridSearchCV with 5-Fold Cross Validation to improve model generalization and reduce overfitting. The datasets were divided into training and testing subsets using an 80:20 split ratio. A fixed random state (42) was used to ensure reproducibility of the experimental results. The training data were used for model training and hyperparameter optimization, while the testing data were reserved for final model evaluation.

3.5 Machine Learning Models

The following regression algorithms were implemented: Linear Regression, Decision Tree Regressor, Random Forest Regressor, Support Vector Regressor (SVR), K-Nearest Neighbors Regressor (KNN), and Gradient Boosting Regressor. Each model was trained and optimized independently.

3.6 Hyperparameter Optimization

Hyperparameter optimization was performed using GridSearchCV with 5-Fold Cross Validation. The coefficient of determination (R^2) was used as the optimization metric. Different parameter combinations were evaluated for each machine learning algorithm in order to improve model generalization performance and reduce overfitting.

For Decision Tree models, maximum tree depth and minimum sample split values were optimized. Random Forest models were tuned using different numbers of estimators, tree depths, and minimum split values. Support Vector Regression models were evaluated with multiple regularization parameters and kernel functions. K-Nearest Neighbors models were optimized using different neighborhood sizes and weighting schemes. Gradient Boosting models were tuned by adjusting the number of estimators, learning rates, and tree depths.

Table 3. Hyperparameter Search Space

Model	Hyperparameters
Decision Tree	max_depth = {None, 5, 10, 20}, min_samples_split = {2, 5, 10}
Random Forest	n_estimators = {50,100}, max_depth = {None,10,20}, min_samples_split = {2,5}
SVR	C = {0.1,1,10}, kernel = {linear, rbf}
KNN	n_neighbors = {3,5,7,10}, weights = {uniform, distance}
Gradient Boosting	n_estimators = {50,100}, learning_rate = {0.01,0.1,0.2}, max_depth = {3,5}

Table 4. Best Hyperparameters

Dataset	Best Model	Best Parameters
Gaming	Gradient Boosting	learning_rate=0.1, max_depth=5, n_estimators=100
Mental	Random Forest	max_depth=10, min_samples_split=5, n_estimators=100
Student	Gradient Boosting	learning_rate=0.1, max_depth=3, n_estimators=100

3.7 Model Evaluation

Model performance was evaluated using RMSE, MAE, R^2 Score, and Adjusted R^2 Score to assess prediction accuracy and generalization. In addition to Feature Importance and SHAP analysis, Permutation Importance was applied to measure feature relevance by observing performance changes after randomly shuffling feature values. The results were used to validate the consistency of feature importance rankings obtained from other explainability methods.

3.8 Explainability Analysis

Explainability techniques were applied to understand how individual variables influenced model predictions.

Feature Importance. Feature Importance was applied to tree-based models such as Random Forest, Decision Tree, and Gradient Boosting.

Permutation Importance. Permutation Importance was used for models without built-in feature importance mechanisms.

SHAP Analysis. SHAP (SHapley Additive Explanations) was used to provide detailed explanations of feature contributions and model behavior.

4 Results

4.1 Model Performance Comparison

The performance of all regression models was compared using RMSE, MAE, R^2 , and Adjusted R^2 metrics. Tables 5, 6, and 7 summarize the results for the Gaming, Mental Health, and Student datasets, respectively. The best-performing model in each dataset is highlighted in bold.

Table 5. Gaming Dataset Model Results.

Model	RMSE	MAE	R^2	Adjusted R^2
Linear Regression	6.5632	5.2327	0.9141	0.9129
Decision Tree	7.7472	6.0284	0.8803	0.8786
Random Forest	6.1882	4.7826	0.9236	0.9225
SVR	6.2970	4.9120	0.9209	0.9198
KNN	8.7623	6.8433	0.8468	0.8447
Gradient Boosting	5.9936	4.6847	0.9283	0.9273

Table 6. Mental Health Dataset Model Results.

Model	RMSE	MAE	R^2	Adjusted R^2
Linear Regression	0.7366	0.5895	-0.4282	–
Decision Tree	0.6209	0.3651	-0.0146	–
Random Forest	0.6088	0.3907	0.0245	–
SVR	0.6455	0.3963	-0.0966	–
KNN	0.6588	0.4148	-0.1424	–
Gradient Boosting	0.6123	0.3821	0.0132	–

Table 7. Student Dataset Model Results.

Model	RMSE	MAE	R^2	Adjusted R^2
Linear Regression	0.1588	0.1286	0.9121	0.9106
Decision Tree	0.1580	0.1209	0.9129	0.9114
Random Forest	0.1297	0.1045	0.9413	0.9403
SVR	0.1434	0.1170	0.9283	0.9270
KNN	0.2216	0.1747	0.8288	0.8259
Gradient Boosting	0.1172	0.0976	0.9521	0.9513

The comparison results indicate that ensemble-based machine learning methods generally achieved superior predictive performance compared to simpler baseline models. For the Gaming Dataset, Gradient Boosting achieved the highest R^2 score (0.9283) and the lowest RMSE value, demonstrating strong predictive capability. For the Student Dataset, Gradient Boosting again produced the best overall results with an R^2 score of 0.9521, indicating that academic and behavioral variables are highly informative for predicting student success. In contrast, the Mental Health Dataset produced considerably lower predictive performance. Random Forest achieved the highest R^2 score (0.0245), suggesting that the available mental health variables alone were insufficient to explain variations in academic performance. Overall, the results show that educational and behavioral factors provide stronger predictive signals than psychological indicators in the examined datasets.

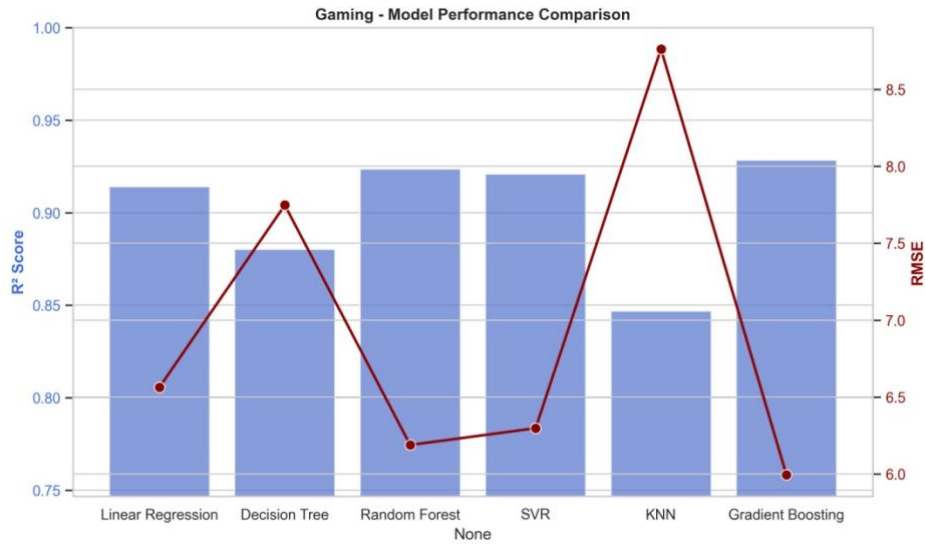


Fig. 6. Gaming Dataset Model Comparison.

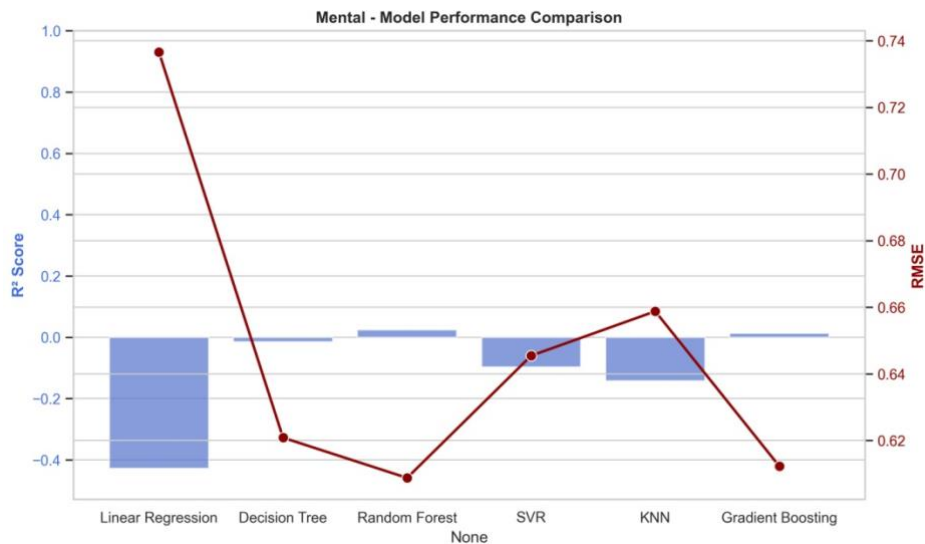


Fig. 7. Mental Health Dataset Model Comparison.

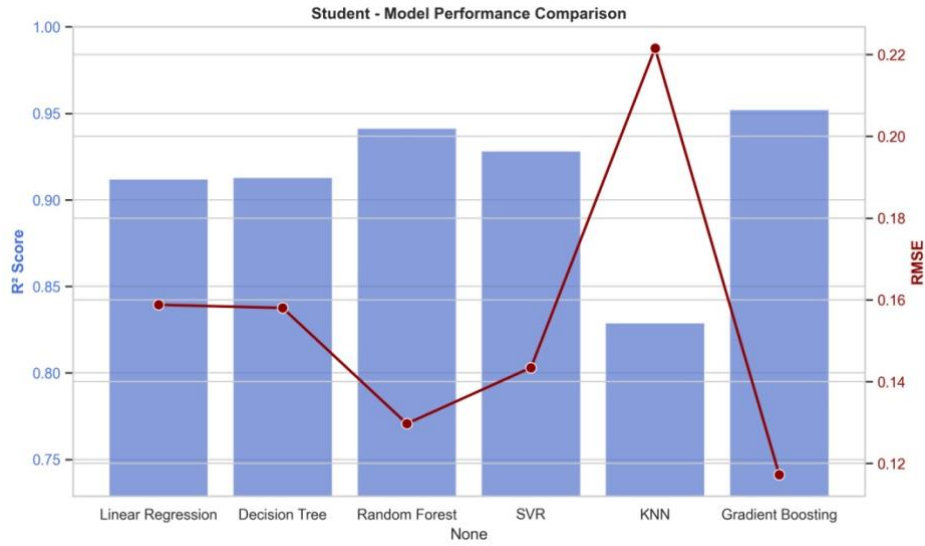


Fig. 8. Student Dataset Model Comparison.

4.2 Prediction Performance

Actual versus predicted visualizations were generated to evaluate how closely the models predicted the target variables. Figures 9, 10, and 11 show these plots for the Gaming, Mental Health, and Student datasets, respectively.

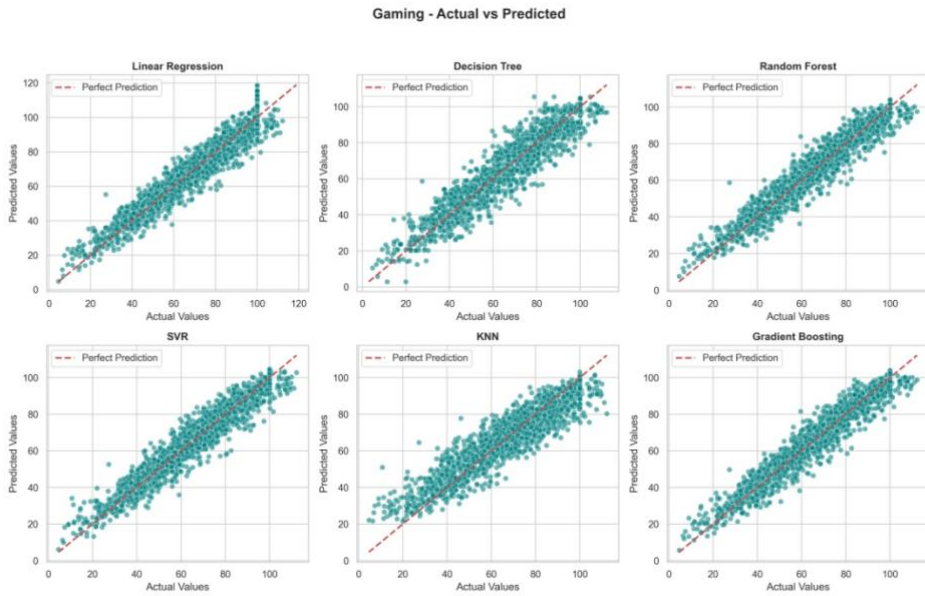


Fig. 9. Gaming Dataset Actual vs Predicted.

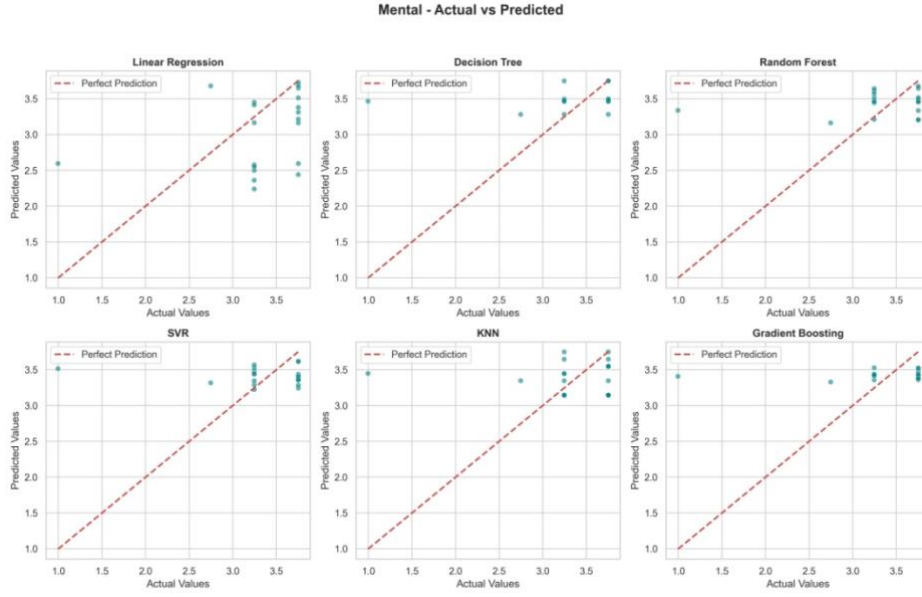


Fig. 10. Mental Health Dataset Actual vs Predicted.

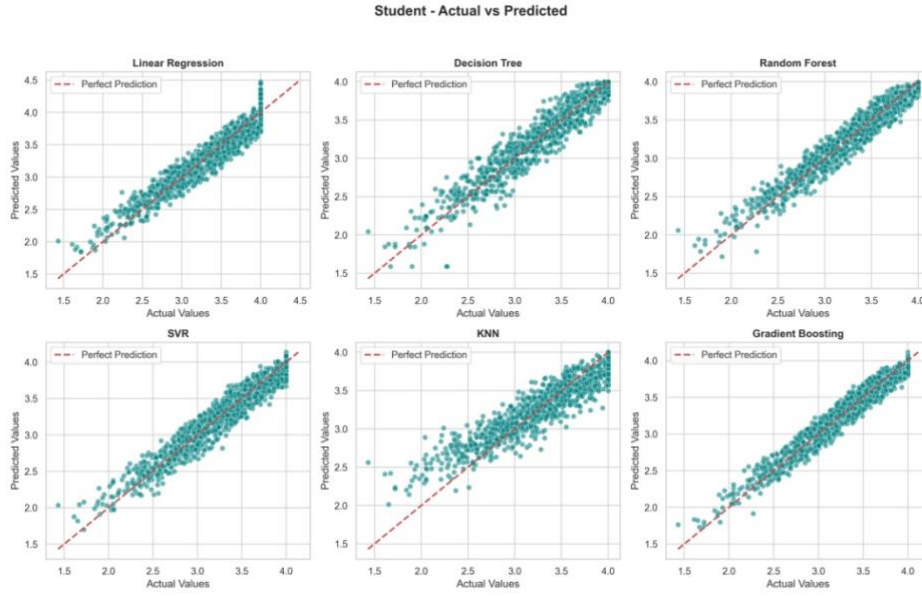


Fig. 11. Student Dataset Actual vs Predicted.

The Actual vs Predicted plots provide a visual assessment of model performance. For the Gaming and Student datasets, the predicted values closely follow the reference line, indicating strong agreement between actual and predicted outcomes. This observation is consistent with the high R^2 scores obtained by the best-performing models. In contrast, the Mental Health dataset exhibits greater dispersion around the reference line, reflecting the lower predictive performance observed during model evaluation. These findings further support the conclusion that academic and behavioral variables are more effective predictors of academic performance than the psychological variables available in the Mental Health dataset.

4.3 Explainability Results

Explainability techniques identified the variables that contributed most strongly to prediction outcomes. Figures 12 to 17 present the Feature Importance and SHAP summary plots for each dataset.

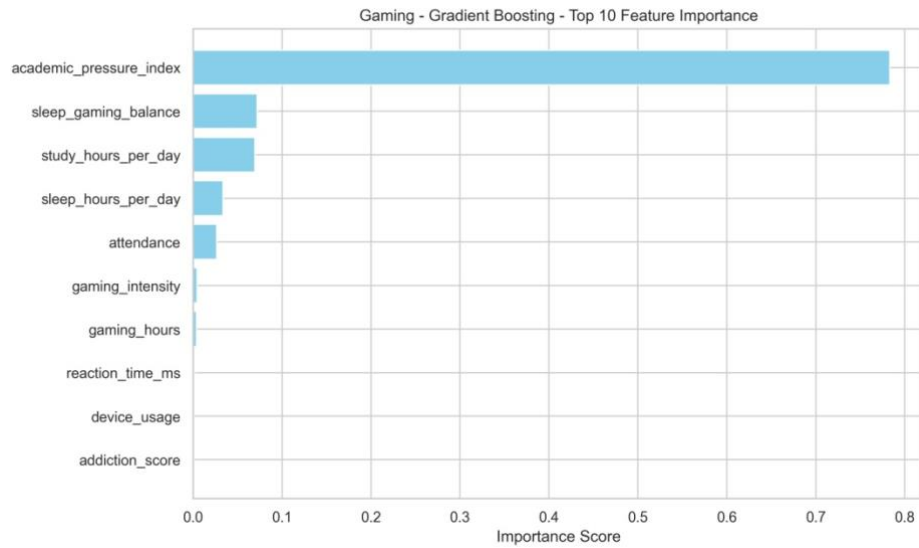


Fig. 12. Gaming Dataset Feature Importance.

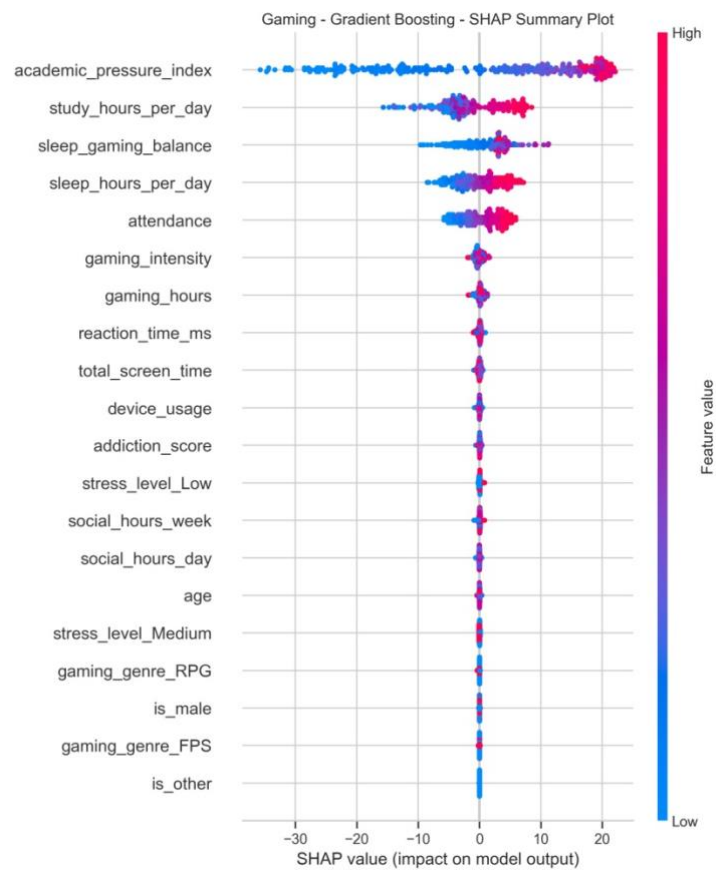


Fig. 13. Gaming Dataset SHAP Summary Plot.

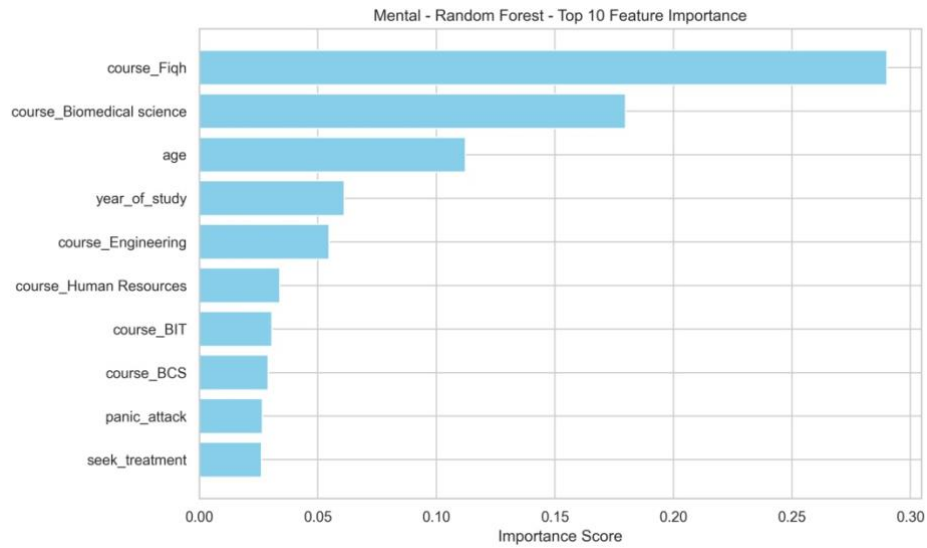


Fig. 14. Mental Health Dataset Feature Importance.

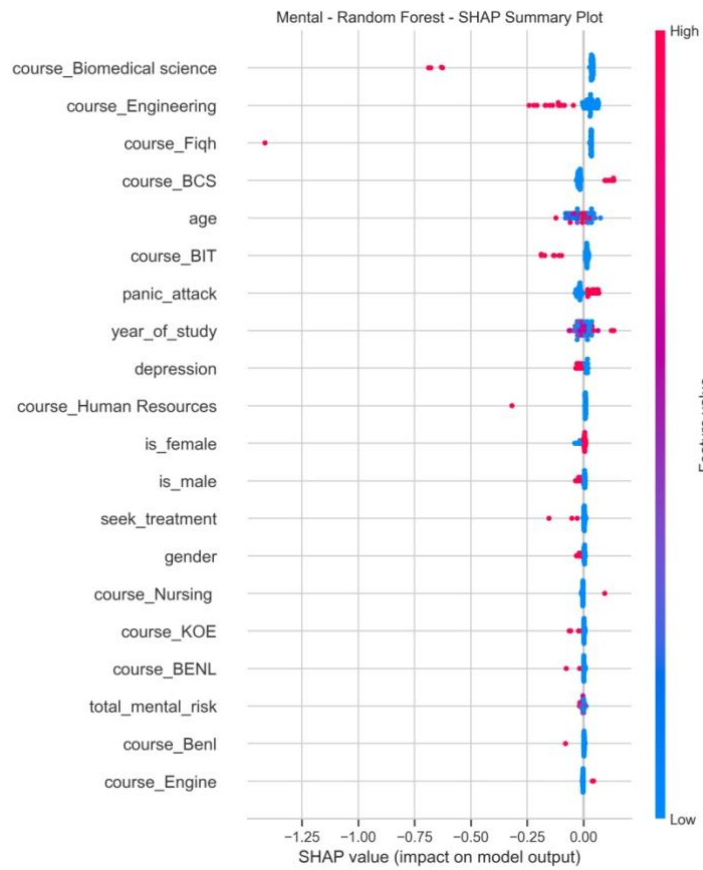


Fig. 15. Mental Health Dataset SHAP Summary Plot.

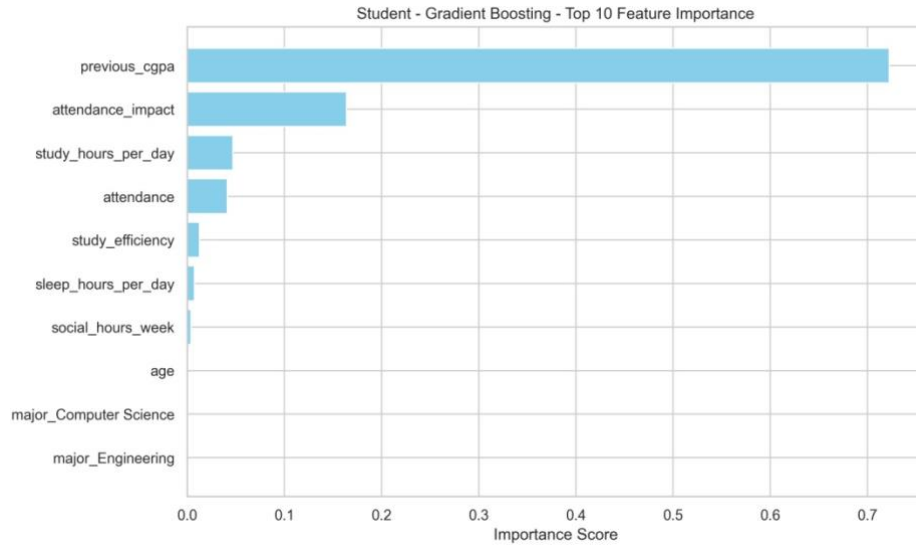


Fig. 16. Student Dataset Feature Importance.

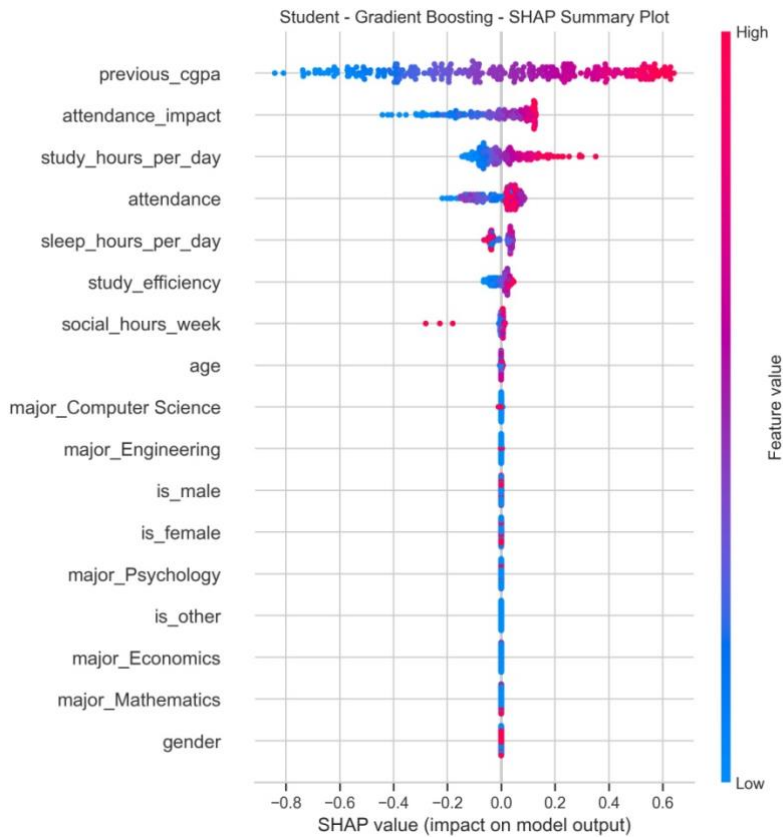


Fig. 17. Student Dataset SHAP Summary Plot.

The explainability analysis provided valuable insights into the factors influencing academic performance predictions. For the Gaming Dataset, Academic Pressure Index emerged as the most influential feature, followed by Sleep-Gaming Balance and Study Hours per Day. In the Student Dataset, Previous CGPA was identified as the dominant predictor, with Attendance Impact and Study Hours per Day also contributing significantly to model predictions. For the Mental Health Dataset, academic background variables showed stronger contributions than mental health indicators. The SHAP analyses confirmed these findings and provided additional information regarding the direction and magnitude of feature effects on model predictions. These results improve the interpretability of the machine learning models and help explain the factors that

influence academic success. These findings are consistent with existing educational research, which emphasizes the importance of previous academic achievement, study habits, and attendance in determining future academic success. The explainability analyses therefore not only improved model transparency but also validated established educational theories using data-driven methods.

5 Discussion

The experimental results demonstrate that educational and behavioral variables provide substantially stronger predictive signals than psychological indicators when predicting academic performance. The Student Dataset and Gaming Dataset achieved considerably higher predictive accuracy compared to the Mental Health Dataset.

One possible explanation is that variables such as attendance, previous academic achievement, study habits, and gaming behavior are directly related to learning outcomes. These variables capture observable behaviors that strongly influence academic success. In contrast, mental health variables may affect academic performance indirectly and therefore require additional contextual information to improve predictive accuracy.

The results also highlight the effectiveness of ensemble learning methods. Random Forest and Gradient Boosting consistently outperformed simpler regression techniques across multiple datasets. This finding suggests that nonlinear relationships exist among academic, behavioral, and psychological variables and can be effectively captured by ensemble-based machine learning approaches.

Furthermore, explainability analyses confirmed that previous academic performance, attendance, study habits, and behavioral indicators were among the most influential predictors. These findings are consistent with previous Educational Data Mining studies that emphasize the importance of academic engagement and learning behavior in determining student success.

The findings obtained in this study are consistent with previous Educational Data Mining research, which identified attendance, previous academic achievement, and study habits as major determinants of academic success. Similar studies have also reported that ensemble learning methods frequently outperform traditional regression models due to their ability to capture nonlinear relationships among educational variables. Among all evaluated datasets, the Student Dataset achieved the highest predictive performance, reaching an R^2 score of 0.9521 with the Gradient Boosting model. This result indicates that academic and behavioral variables provide highly informative signals for predicting future academic achievement. These findings suggest that behavioral and educational variables provide more reliable signals for academic performance prediction than psychological indicators alone. The results also demonstrate the practical value of explainable machine learning approaches in educational analytics.

6 Quality Assurance and Testing

An automated testing framework was developed to ensure the reliability and reproducibility of the project pipeline. The testing process covered multiple stages, including raw dataset validation, cleaned dataset verification, feature engineering validation, machine learning pipeline testing, evaluation metric verification, generated report validation, and dependency checks. These tests helped ensure that each component of the project operated correctly and produced consistent outputs.

The testing process also included verification of generated visualizations, validation of saved report files, and consistency checks across different datasets. Automated tests

were executed throughout the development process to identify potential errors and ensure reproducibility of experimental results.

7 Threats to Validity

Several limitations should be considered when interpreting the results of this study. First, the datasets were obtained from publicly available sources and may not fully represent all student populations or educational environments. Different institutions, countries, and educational systems may exhibit different behavioral patterns and academic outcomes. Second, some potentially important variables such as socioeconomic background, family support, learning styles, and institutional characteristics were not available in the datasets. The absence of these variables may limit the predictive capability of the developed models. Third, although GridSearchCV and 5-Fold Cross Validation were used to improve model generalization, performance may vary when the models are applied to unseen populations. In addition, the study focused on traditional machine learning algorithms and did not evaluate deep learning architectures or advanced ensemble stacking approaches. Future studies should address these limitations by incorporating larger datasets, additional explanatory variables, and more sophisticated predictive techniques.

8 Conclusion

This project developed a complete machine learning framework for predicting academic performance using student, gaming, and mental health datasets. The workflow included data preprocessing, feature engineering, exploratory data analysis, model training, evaluation, explainability analysis, and automated testing. The results demonstrated that machine learning models can effectively predict academic outcomes when educational and behavioral variables are available. The Student and Gaming datasets achieved strong predictive performance, while the Mental Health dataset highlighted the challenges of modeling academic success using psychological indicators alone. Furthermore, explainability techniques such as Feature Importance, Permutation Importance, and SHAP provided valuable insights into the factors that most strongly influence academic performance. Future work may include larger datasets, additional features, and advanced deep learning approaches to further improve predictive accuracy and model interpretability. The proposed framework demonstrates how machine learning and explainable artificial intelligence can be combined to support evidence-based educational decision making. The results highlight the potential of data-driven approaches for understanding and improving student academic outcomes. The findings of this study demonstrate the practical value of combining machine learning and explainable artificial intelligence techniques within educational analytics. Beyond prediction accuracy, the proposed framework provides interpretable insights that can support educators, institutions, and policymakers in developing evidence-based strategies to improve student success.

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